Ethan Labbe

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WORK EXPERIENCE

Controls Engineer Intern Jan 2025 – Aug 2025
Tesla — Elgin, IL, USA

- Led end-to-end development of an **automated vision-inspection** station, validating stator geometry within a **17-second cycle time.**
- Developed computer-vision algorithms to analyze **2D line-scan imagery** and **3D profilometer point clouds**, measuring stator surface features and validating dimensional and geometric requirements.
- Calibrated, tested, and validated multiple vision systems that measured stator features with a maximum **200 µm error** to accurately identify component faults.
- Programmed PLC motion-control logic for 50+ devices to synchronize **multi-axis industrial robots**, servo motors, conveyors, and a dial table using ladder logic, Structured Text, and sequential graphs.
- Engineered TCP/IP communications to synchronize PLC control logic with IPC vision systems.
- Configured dynamic HMIs in Siemens TIA Portal for manual control, alarm handling, and diagnostics.
- Configured and integrated Siemens safety-rated I/O devices with fail-safe PLC logic to ensure strict compliance with industrial machine safety standards.

Controls Engineer Intern Jan 2024 – Aug 2024
General Fusion — Vancouver, BC, Canada

- Designed and deployed a **Python-based data acquisition** system for **fusion reactor diagnostics.**
- Developed Python instrument drivers to control oscilloscopes, digitizers, and cameras.
- Automated **TCP/IP communications** between LabVIEW control-system interfaces and DAQ devices.
- Programmed **Structured-Text PLC code** in TwinCAT to interface with Python DAQ systems via TCP/IP.
- Leveraged Git version control to maintain clean modular code and streamline collaboration.

Controls Systems Integrator May 2023 – Sep 2023
Cybertech Automation — Edmonton, AB, Canada

- Led the design, installation and documentation of a power monitoring system to accurately measure power consumption of a three-phase MCC panel.
- Engineered a control subpanel to capture and communicate energy data via Modbus TCP protocols.
- Configured, tested, and deployed multiple custom **SCADA HMI interfaces**.

PROJECT EXPERIENCE

ROS 2 Edge-Computing Architecture for Robotic Inspection Jun 2026 – Present

- Developing a **ROS 2 edge pipeline** that offloads inspection workloads from an onboard Raspberry Pi 4 to an external processor when anomalous conditions or resource limits are detected.
- Building an **Ignition Gazebo hardware-in-the-loop simulation environment** to evaluate real-time inspection-pipeline performance using a Raspberry Pi 4 and camera hardware.

Handheld Stereo Vision Depth Mapping System Sept 2025 – April 2026

- Designed and built a low-cost handheld stereo-vision device integrating a Raspberry Pi 5, AI cameras, a touchscreen display, and an external viewer/processor
- Developed a Python/OpenCV vision pipeline for synchronized camera acquisition, stereo calibration, depth-map generation, and wireless image transfer.
- Calibrated, tested and validated the system to produce 3D depth maps with **2.5 mm accuracy.**

Inertial Fusion Energy Fuel Research Project Sept 2023 - Dec 2023

- Designed and prepared laser-scattering experiments at the University of Alberta Laser Laboratory to evaluate inertial-fusion fuel targets.
- Captured and analyzed camera-based laser-scattering measurements to support fuel-target characterization and manufacturing-cost reduction.

TECHNICAL SKILLS

Programming: Python, C/C++, MATLAB, Structured Text, Ladder Logic, Linux, Docker
Vision & Robotics: OpenCV, HALCON, ROS 2, Ignition Gazebo, camera calibration, vision system integration, line-scan imaging, area-scan imaging, 3D point-cloud processing, sensor fusion
Controls & Integration: Siemens TIA Portal, TwinCAT, PLCs, HMI/SCADA, TCP/IP, Safety I/O, Modbus TCP, DAQ, Git/GitHub, CI/CD, KiCad, AutoCAD, circuit design, hardware–software integration

EDUCATION

Electrical Engineering, BSc Co-op, University of Alberta — Class of 2026
Cumulative Grade Point Average — 3.8/4.0